

Targeted therapy for thyroid cancers

Targeted therapy drugs interfere with the way that cancer cells grow. They are sometimes used to treat people with thyroid cancer.

On this page

- [About targeted therapy for thyroid cancers](#)
- [Targeted therapy for papillary or follicular thyroid cancer](#)
- [Targeted therapy for anaplastic thyroid cancer](#)
- [Targeted therapy for medullary thyroid cancer](#)
- [Side effects of targeted therapy for thyroid cancer](#)
- [Get in touch](#)

About targeted therapy for thyroid cancers

[Targeted therapy drugs](#) target something in or around the cancer cell that is helping it grow and survive. They may be used to treat some types of [thyroid cancer](#).

You take the drugs as either capsules or tablets every day. Always take them exactly as your healthcare team tell you to.

Sometimes targeted therapy drugs are given as part of a [clinical trial](#). Your cancer doctor will give you more information about these drugs if they are available and suitable for you.

Targeted therapy for papillary or follicular thyroid cancer

Targeted therapies might be used if papillary or follicular thyroid cancer has:

- stopped responding to radioactive iodine treatment.
- spread to other parts of the body
- come back.

Lenvatinib and sorafenib

The 2 main targeted therapy drugs used to treat [papillary or follicular thyroid cancer](#) are:

- [lenvatinib](#) (Lenvima[®]) – capsules taken once a day
- [sorafenib](#) (Nexavar[®]) – tablets taken 2 times a day.

Lenvatinib and sorafenib are a type of targeted drugs called tyrosine kinase inhibitors (TKIs). They block (inhibit) signals in the cancer cells that make them grow and divide. They can help to shrink the cancer and to control it, sometimes for a long time.

Gene changes

Other targeted therapy drugs may be used to treat thyroid cancer if there is a certain gene change (mutation) in the cancer cells. For example, they may work if the cancer cells have changes in the RET, NTRK, or BRAF gene.

Your cancer doctor may test cancer cells taken from your [biopsy](#) or [thyroid surgery](#) to look for these gene changes. This is called molecular testing.

If the cancer cells have a change in the RET gene your doctor might talk to you about having a drug called [selpercatinib](#) (Retsevmo[®]). This drug belongs to a group of targeted therapy drugs known as [cancer growth inhibitors](#). It can only be used if the thyroid cancer cells have a change in a gene called RET.

This gene change encourages the cancer cells to grow and divide. Selpercatinib blocks the effect of this gene change and slows or stops the cancer cell growth.

You might have selpercatinib either:

- if lenvatinib or sorafenib are not controlling the cancer
- as your first targeted therapy drug if radioactive iodine has not got rid of all the cancer.

You take selpercatinib as capsules, 2 times a day.

Your cancer doctor will discuss with you if any of these drugs are suitable and available for you.

Targeted therapy for anaplastic thyroid cancer

Targeted therapies are often the main treatment for [anaplastic thyroid cancer](#).

If the cancer cells have a gene change (mutation) called a BRAF V600E mutation, the targeted therapy drugs [dabrafenib and trametinib](#) may be given together. Dabrafenib is taken as capsules and trametinib as tablets.

Some people might have a targeted therapy drug called [lenvatinib](#) along with an immunotherapy drug called [pembrolizumab](#). You take lenvatinib as capsules. You have pembrolizumab as a drip into a vein (infusion).

Targeted therapy for medullary thyroid cancer

These drugs cannot cure [medullary thyroid cancer \(MTC\)](#), but they can slow its growth and help with symptoms.

[Cabozantinib](#) (Cometriq[®]) and vandetanib (Caprelsa[®]) are types of targeted therapy drugs called tyrosine kinase inhibitors (TKI). They work by blocking signals that encourage the thyroid cancer cells to grow and divide.

Cabozantinib or vandetanib are used to treat medullary thyroid cancer when:

- you cannot have surgery
- the cancer is causing symptoms
- the cancer has spread.

Vandetanib is not widely available throughout the UK. If you and your cancer doctor agree that you would benefit from the treatment, they can apply to the NHS for funding. We have more information about [what you can do if a treatment is not available](#).

Gene changes

There are also other targeted therapies that work if the medullary cancer cells have gene changes (mutations) called RET mutations.

[Selpercatinib](#) (Retsevmo[®]) belongs to a group of targeted therapy drugs known as cancer growth inhibitors. It can only be used to treat MTC if the cancer cells have a change in a gene called RET.

The gene change encourages the cancer cells to grow and divide. Selpercatinib blocks the effect of this gene change and slows or stops the cancer cell growth.

You take selpercatinib as capsules, 2 times a day.

Side effects of targeted therapy for thyroid cancer

Your cancer doctor or specialist nurse will talk to you about the possible side effects of [targeted therapy drugs](#).

They will give you advice about how side effects can be managed and treated. Always tell them about any side effects you have and follow their advice.

We also have information about side effects in our [A-Z drug information](#).

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Below is a sample of the sources used in our thyroid cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Get in touch



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.